Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

EIN 4360 Facilities Planning and Material Handling

Module 11 – Evaluating and Selecting a Plan: Break Even Analysis

Nicolas Grimaldi

E-mail: n.grimaldi@ufl.edu

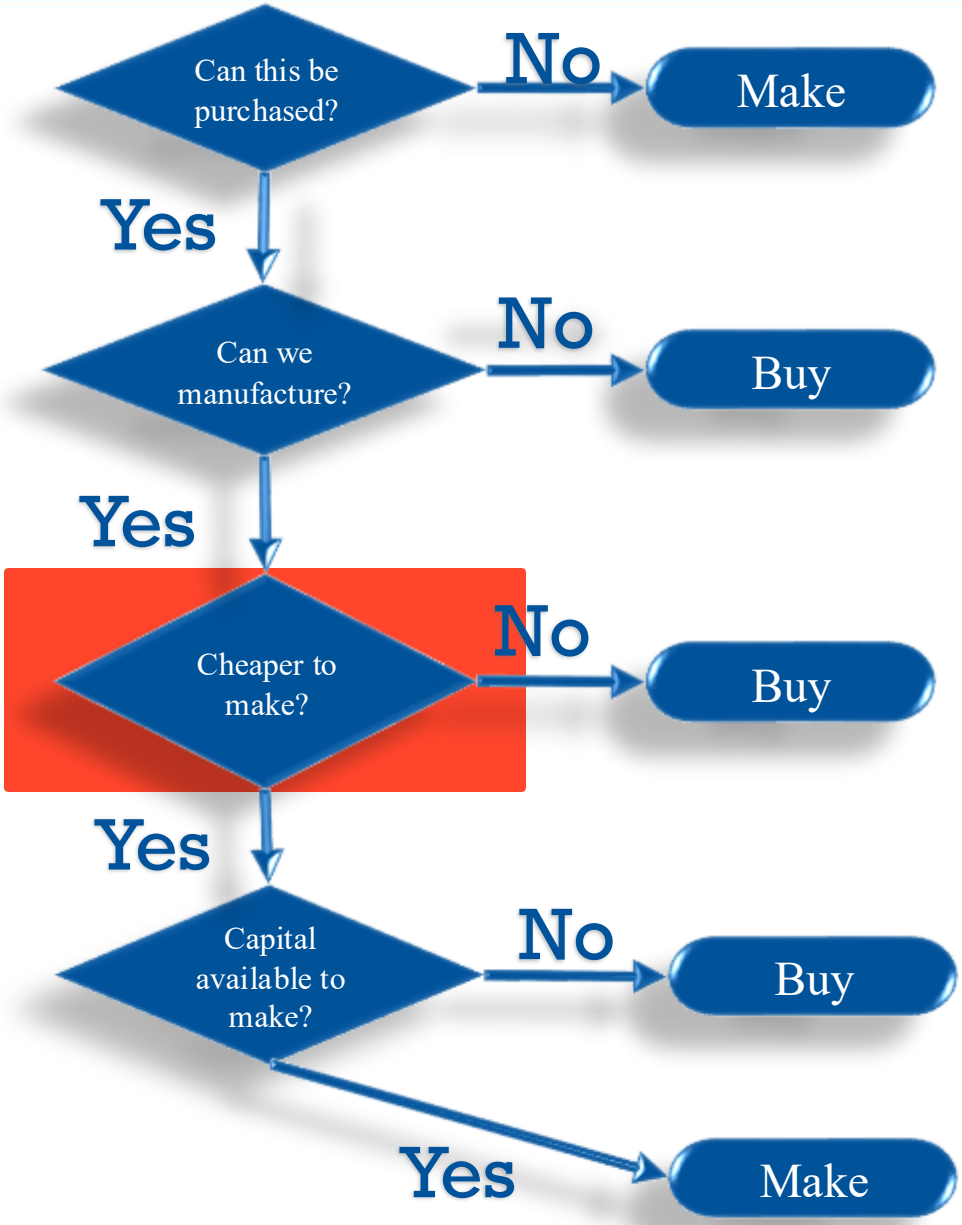
Office: Weil 401

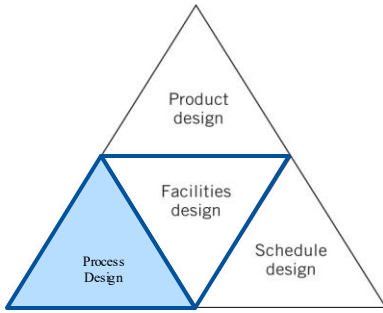
POWERING THE NEW ENGINEER TO TRANSFORM THE FUTURE

Methods for Evaluating Facility Plans

- List **positive** and **negative** aspects of each alternative.
- **Rank** the performance of each alternative against criteria.
- Perform **weighted factor comparison** of the alternatives
- Analytic Hierarchy Process (**AHP**)
- Determine the economic performance of each alternative over a specified planning horizon

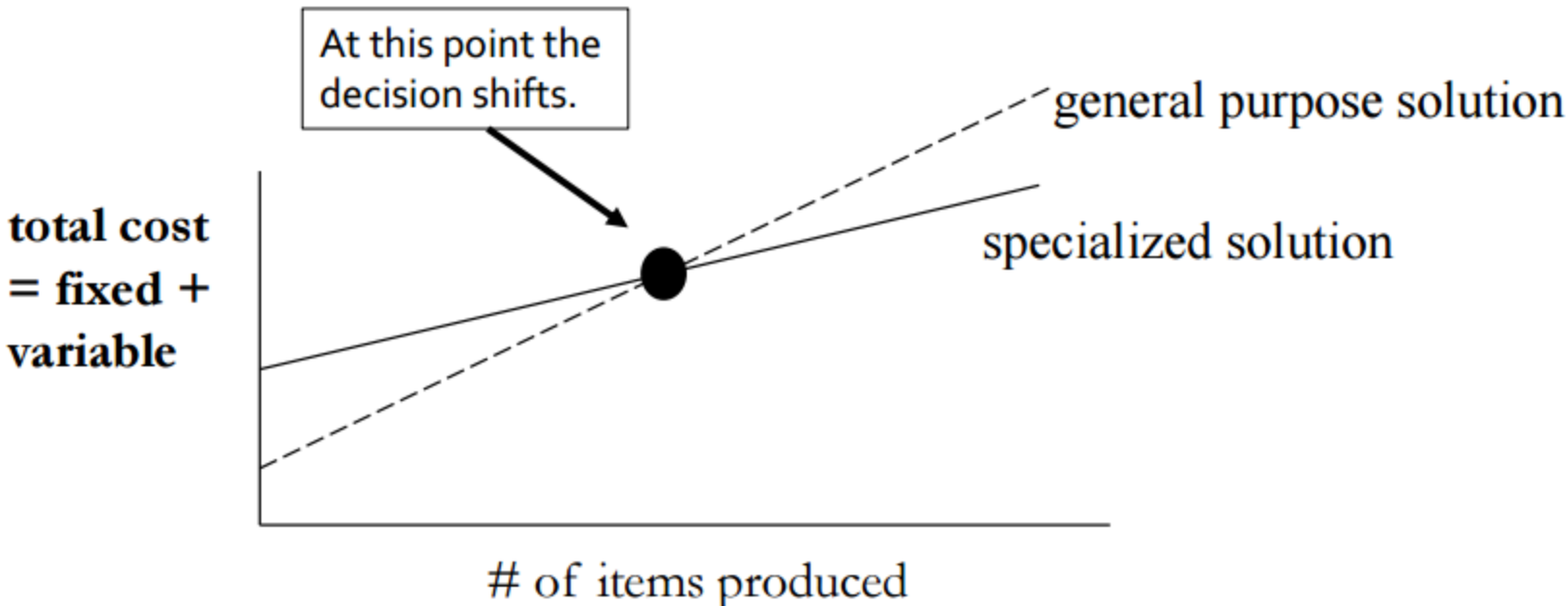
Make vs. Buy


Can this be purchased?
No
Make
Yes
Can we manufacture?
No
Buy
Yes
Cheaper to make?
No
Buy
Yes
Capital available to make?
No
Buy
Yes
Make


Product design
Facilities design
Process Design
Schedule design

Crossover Chart

- aka. **Break-even chart**
- Compare two alternatives - usually general-purpose v. specialized
- Recall: make vs. buy decisions
- Based on consideration of production rates



Fixed vs. Variable Costs

- **Variable Costs:** costs that vary with the amount of usage. These are often costs associated with the amount of goods or services a company/facility produces. Increases and decreases with production volume.
  - *Examples: cost of materials, cost of utilities, cost of shipping, etc.*

- **Fixed Costs:** costs incurred by a company/facility that does not vary with the volume of production.
  - *Examples: costs of purchasing the machine, rent, etc.*

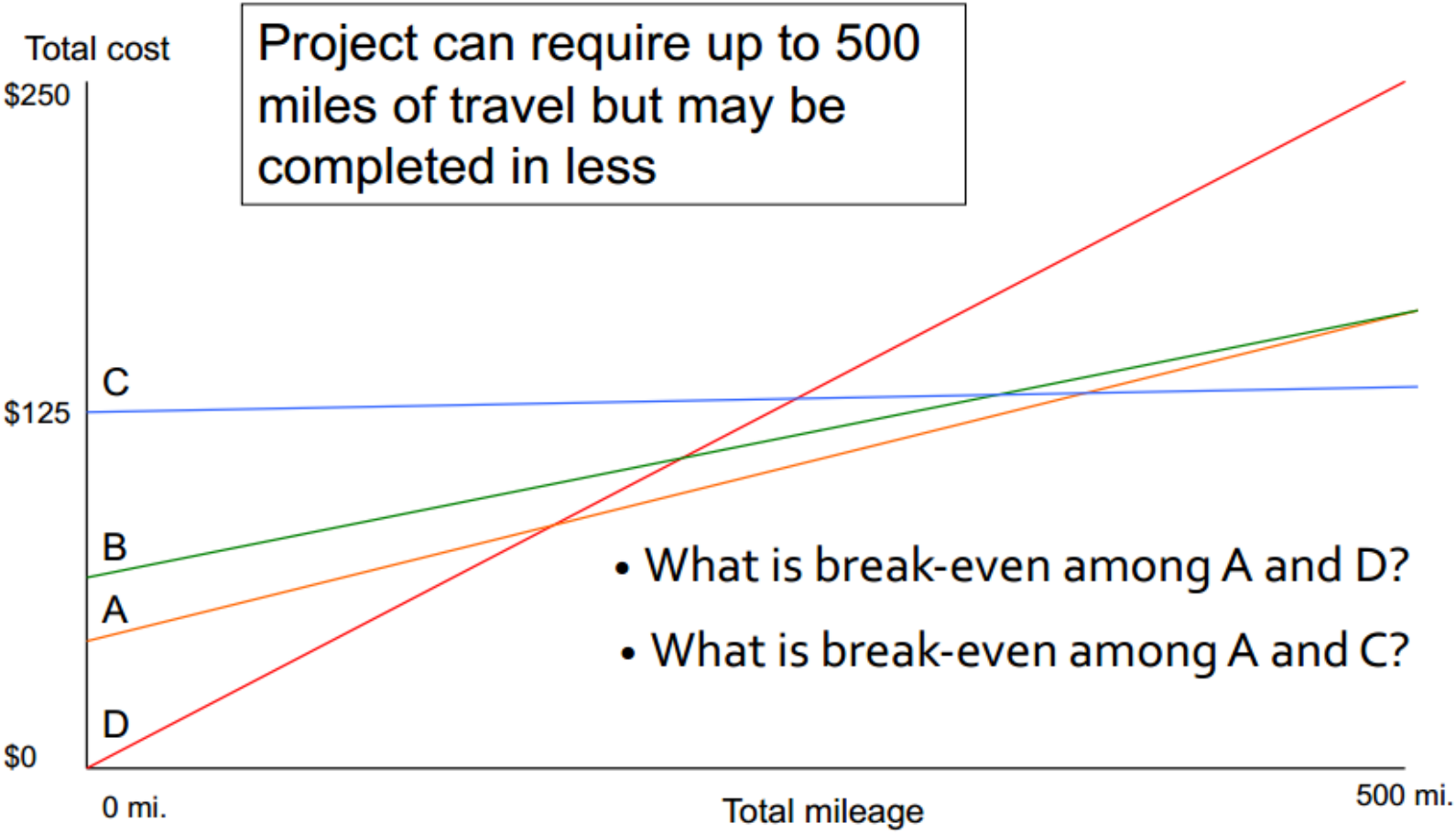
Rental Car Example

- 4 alternate modes of transportation:
  - A = \$50/day + \$0.20/mi.
  - B = \$75/day + \$0.15/mi.
  - C = \$125/day + \$0.01/mi.
  - D = \$0/day + \$0.50/mi.

- Want to travel up to 500 miles in one day. Where are break-even points among alternatives?

- Determine the start and end points:
  - A = 50 ; \$50 + \$0.2(500) = 150
  - B = 75 ; \$75 + \$0.15(500) = 150
  - C = 125 ; \$125 + \$0.01(500) = 130
  - D = 0 ; \$0 + \$0.5(500) = 250

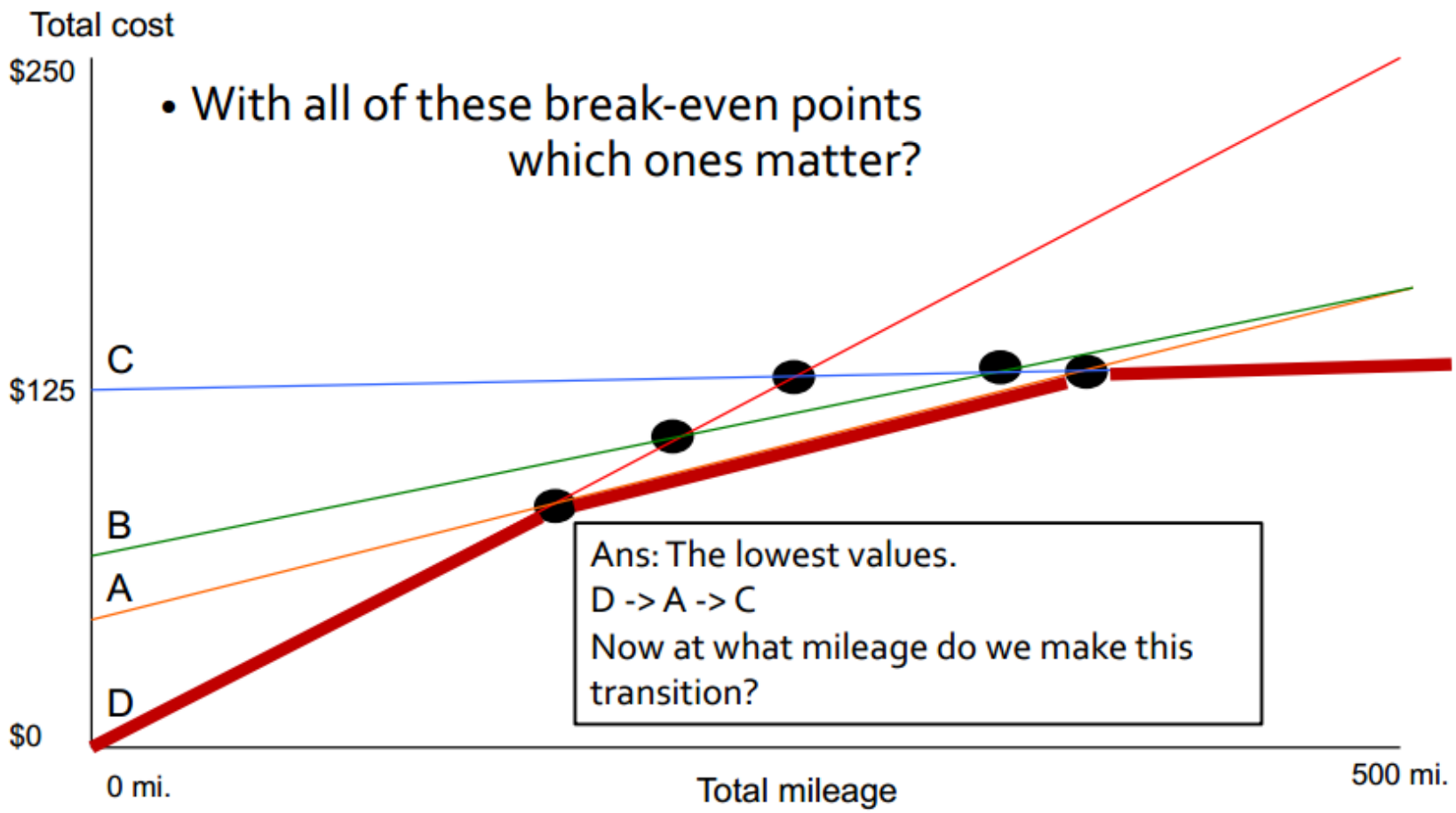
Rental Car Example



Project can require up to 500 miles of travel but may be completed in less

- What is break-even among A and D?
- What is break-even among A and C?

Rental Car Example



Rental Car Example

$$TC_A = 50 + 0.2X$$
$$TC_B = 75 + 0.15X$$
$$TC_C = 125 + 0.01X$$
$$TC_D = 0 + 0.5X$$

Transition Option D to A:

$TC_D = TC_A$

$0.5X = 50 + 0.2X$

$X = 50/0.3 = 166.7$ miles

Transition Option A to C:

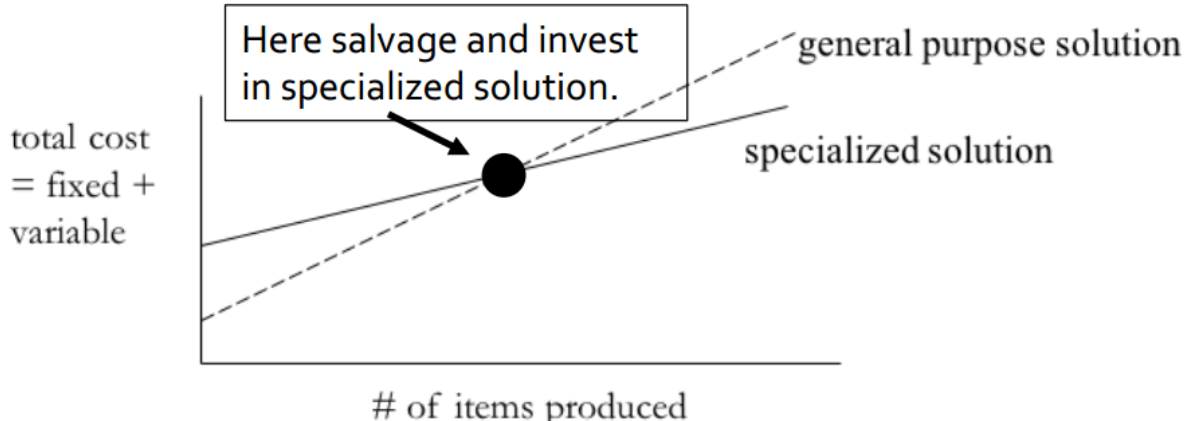
$TC_A = TC_C$

$50 + 0.2X = 125 + 0.01X$

$X = 75/0.19 = 394.7$ miles

Note on Salvage Value

- If investing in equipment, and equipment has salvage value, consider using multiple alternatives during production process life.
  - Begin with general purpose solution.
  - Use until point of cost ineffectiveness (relative to specialized machine).
  - Salvage and use income to offset investment in specialized solution.
  - Need to ensure total cost of combined solution is less than cost of single alternative.



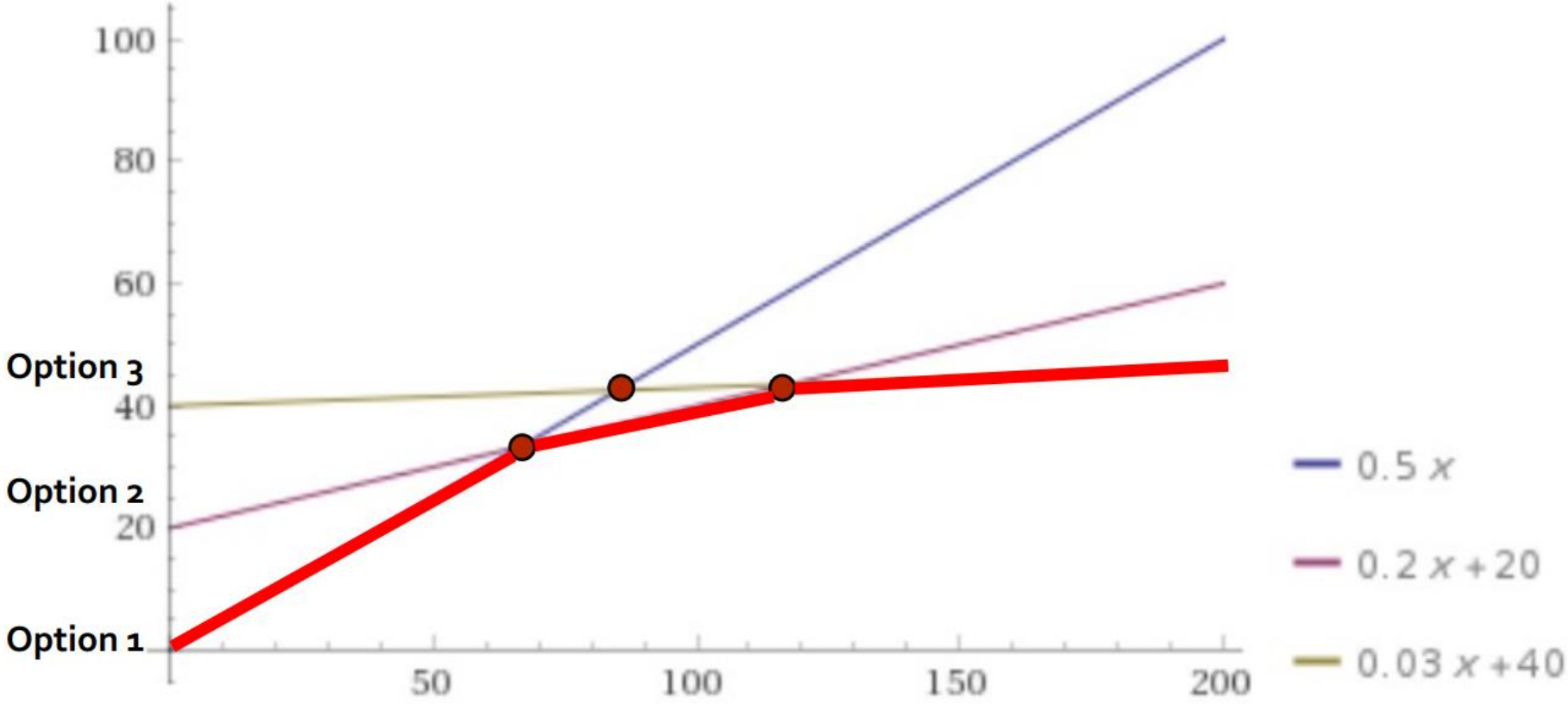
Another Rental Car Example

3 alternate modes of transportation:

- 1 = \$0/day + \$0.50/mi.
- 2 = \$20/day + \$0.20/mi.
- 3 = \$40/day + \$0.03/mi.

1. Write the total cost equations for each
   - TC_1 = _____
   - TC_2 = _____
   - TC_3 = _____
2. Which option is the best for different mileages?

Another Rental Car Example



Option 1 -> Option 2, Option 2 -> Option 3

Another Rental Car Example

1.
   - $TC_1 = 0 + 0.5X$
   - $TC_2 = 20 + 0.2X$
   - $TC_3 = 40 + 0.03X$

2.

Transition Option 1 to 2:

$TC_1 = TC_2$

$0.5X = 20 + 0.2X$

$X = 20/0.3 = 66.7$ miles

Transition Option 2 to 3:

$TC_2 = TC_3$

$20 + 0.2X = 40 + 0.03X$

$X = 20/0.17 = 117.6$ miles

Another Rental Car Example

Taking it one step further:

Write the decision rules for different rental options

- Option 1: <= 66.7 mi.
- Option 2: 66.7 mi. < x <= 117.6 mi.
- Option 3: >117.6 mi.

Economic Decision Tools

- **Return on Investment** (ROI)
  - Use to evaluate the efficiency of an investment or to compare multiple alternatives in terms of their efficiencies.
  - **Ratio of return (benefit)** to **original investment (cost)**

$$ROI = \frac{Gain\ from\ Investment - Cost\ of\ Invesment}{Cost\ of\ Investment} = \frac{Net\ Profit}{Cost\ of\ Investment}$$

  - Choose the alternative that balances the lowest initial investment and greatest gain

Economic Decision Tools

- **Payback Period** (PBP)
  - Reveals the amount of **time to recover capital investment** you made in new alternative method.
  - To calculate the number of years: cost of investment divided by the profits per year.

Economic Decision Tools

- **Discounted Cash Flow**
  - Identifies future cash flow (e.g., profits and costs) associated with new investment and applies a discount to these future cash flows due to the time value of money
  - Determine present value of profits and consider interest earning potential of cash over time.

Time Value of Money

- There is greater benefit of receiving money now rather than an identical sum later.
- Reason being: You might be able to invest the money elsewhere during that time.
- Important when considering different facilities plans and investment options as the benefits and costs of these choices can occur over a long period of time.
- **Planning Horizon:** Defines the width of a window through which we look in evaluating each investment alternative.
  - US firms use too small of a planning horizon (often less than 5 years) compared to Japanese firms that think in terms of 10 to 20 years.

How to Evaluate Alternatives?

- The evaluation of the best financial decision would be the alternative that results in **highest profit, ROI or lowest cost or PBP** after taking into account the following things over the a specified **planning horizon:**
  - The initial cost of the systems (purchase cost, sales tax, cost of delivery, cost of assembly, etc.)
  - Annual operating costs (training, maintenance, etc.)
  - Annual profits (will depend on productivity)
  - Expected useful life
  - Expected salvage value
  - Etc.

Economic Decision Tools

- **Total Present Value** (TPV)
  - Present Value is based on estimates of cash flow and expected interest rate.

$$Present\ Value = \frac{Future\ Value}{(1+i)^n}$$

  - *Future Value* is the future value of an investment
  - *i* is the discount rate (e.g. interest rate, rate of return)
  - *n* is the number of periods (e.g. years)

- **TPV = ΣPV** across all years
- **Net Present Value** (NPV) = TPV – Initial investment cost

Discount Rates

- **Interest Rate:** a risk-free rate of return (e.g. treasury bonds)
- **Required Rate of Return:** a desired return for an investment
- **Weighted Average Cost of Capital:** the average interest rate a company must pay to finance its assets, growth, and work capital. The WACC is the minimum average rate of return it must earn on its current assets to satisfy its shareholders, investors, or creditors.

Economic Decision Tools Example

- Capital Investment for New Machine = $25,000
- Assume interest rate is 8%
- Estimated life of New Machine = 7 years, no salvage value
- Productivity will increase by 20% creating the following data:

Year	Increase in Sales	Increase in Costs
1	$10,000	$1,000
2	$7,000	$700
3	$6,000	$600
4	$5,000	$500
5	$3,000	$300
6	$2,000	$200
7	$2,000	$200

1. Calculate Return on Investment (ROI)
2. Calculate Payback Period (PBP)
3. Calculate Total Present Value (TPV)
4. Calculate Net Present Value (NPV)

Economic Decision Tools Example

- Capital Investment = $25,000
- Interest rate is 8%
- Estimated life = 7 years, no salvage value
- Productivity will increase by 20%

1. Return on Investment = ($31,500-$25,000)/$25,000 = 26%

Year	Increase in Sales	Increase in Costs	Gross Profit
1	$10,000	$1,000	$9,000
2	$7,000	$700	$6,300
3	$6,000	$600	$5,400
4	$5,000	$500	$4,500
5	$3,000	$300	$2,700
6	$2,000	$200	$1,800
7	$2,000	$200	$1,800

Economic Decision Tools Example

- Capital Investment = $25,000
- Interest rate is 8%
- Estimated life = 7 years, no salvage value
- Productivity will increase by 20%

2. Payback Period = 3 + 4,300/4,500 = 3.96 years

Year	Increase in Sales	Increase in Costs	Gross Profit
1	$10,000	$1,000	$9,000
2	$7,000	$700	$6,300
3	$6,000	$600	$5,400
4	$5,000	$500	$4,500
5	$3,000	$300	$2,700
6	$2,000	$200	$1,800
7	$2,000	$200	$1,800

25,000-9,000=16,000

16,000-6,300=9,700

9,700-5,400=4,300

4,300-4,500=-200

Economic Decision Tools Example

- Capital Investment = $25,000
- Interest rate is 8%
- Estimated life = 7 years, no salvage value
- Productivity will increase by 20%

$$Present\ Value = \frac{Future\ Value}{(1+i)^n}$$

3. Total PV over the years = $25,351

Year	Gross Profit	Present Value
1	$9,000	$8,333
2	$6,300	$5,401
3	$5,400	$4,287
4	$4,500	$3,308
5	$2,700	$1,838
6	$1,800	$1,134
7	$1,800	$1,050

- $Year\ 1: Present\ Value = \frac{9{,}000}{(1+0.08)^1} = 8333.33$
- $Year\ 2: Present\ Value = \frac{6{,}300}{(1+0.08)^2} = 5401.23$
- $Year\ 3: Present\ Value = \frac{5{,}400}{(1+0.08)^3} = 4286.69$
- $Year\ 4: Present\ Value = \frac{4{,}500}{(1+0.08)^4} = 3307.63$
- $Year\ 5: Present\ Value = \frac{2{,}700}{(1+0.08)^5} = 1837.57$
- $Year\ 6: Present\ Value = \frac{1{,}800}{(1+0.08)^6} = 1134.31$
- $Year\ 7: Present\ Value = \frac{1{,}800}{(1+0.08)^7} = 1050.28$

Economic Decision Tools Example

- Capital Investment = $25,000
- Interest rate is 8%
- Estimated life = 7 years, no salvage value
- Productivity will increase by 20%

4. Net Present Value = $25,351 - $25,000 = $351

Note: Excel: NPV function -> Actually calculates the total present value! Need to subtract the initial investment.

COUNTIF | =NPV(B2,B6:B8)+B5

	A	B	C	D	E
1	Project X				
2	Discount Rate	15%			
3					
4	Period	Cash Flow			
5	0	-100			
6	1	300			
7	2	500			
8	3	1000			
9					
10	Net Present Value	$1,196.46			
11					

In-class Activity

You are looking at replacing an old drilling machine with a new one. The new machine will cost **$5000** to purchase, and can be salvaged in 6 years for **$750**. The old machine has a salvage value of $0. You have estimated the following cashflows for each alternative. Assume a required rate of return of 5%

	Cashflow	
Year	New Machine	Old Machine
1	-500	2000
2	2500	2000
3	2800	750
4	2800	200
5	2000	100
6	2000	100

1. Calculate Return on Investment (ROI)
2. Calculate Payback Period (PBP)
3. Calculate Net Present Value (NPV)

In-class Activity

Solution

You are looking at replacing an old drilling machine with a new one. The new machine will cost **$5000** to purchase, and can be salvaged in 6 years for **$750**. The old machine has a salvage value of $0. You have estimated the following cashflows for each alternative. Assume a required rate of return of 5%

	Cashflow		
Year	New Machine	Old Machine	Gain from Investment
1	-500	2000	-2500
2	2500	2000	500
3	2800	750	2050
4	2800	200	2600
5	2000	100	1900
6	2000	100	1900

1. Calculate Return on Investment (ROI)

$$ROI = \frac{Gain\ from\ Investment - Cost\ of\ Invesment}{Cost\ of\ Investment}$$

$$ROI = \frac{(6450+750) - 5000}{5000} = 44\%$$

In-class Activity

Solution

You are looking at replacing an old drilling machine with a new one. The new machine will cost **$5000** to purchase, and can be salvaged in 6 years for **$750**. The old machine has a salvage value of $0. You have estimated the following cashflows for each alternative. Assume a required rate of return of 5%

	Cashflow		
Year	New Machine	Old Machine	Gain from Investment
1	-500	2000	-2500
2	2500	2000	500
3	2800	750	2050
4	2800	200	2600
5	2000	100	1900
6	2000	100	1900

2. Calculate Payback Period (PBP)

PBP = 5+450/1,900
= 5.23 years

In-class Activity

Solution

You are looking at replacing an old drilling machine with a new one. The new machine will cost **$5000** to purchase, and can be salvaged in 6 years for **$750**. The old machine has a salvage value of $0. You have estimated the following cashflows for each alternative. Assume a required rate of return of 5%

	Cashflow		
Year	New Machine	Old Machine	Gain from Investment
1	-500	2000	-2500
2	2500	2000	500
3	2800	750	2050
4	2800	200	2600
5	2000	100	1900
6	2000	100	1900

3. Calculate Net Present Value (NPV)
Key: Account for salvage value in year 6

NPV = 5448.63-5000 = $448.63